HOLOMORPHIC DEFINING COORDINATES FOR SYMPLECTIC NORMAL-CROSSINGS DIVISORS

Abstract. The answer is affirmative, with the cohomologous deformation allowed in the question. The proof first enlarges the divisor by auxiliary symplectic hypersurfaces so that their cohomology classes satisfy a redundancy condition. An integral matrix of relations among these classes then produces globally defined, smooth, nonvanishing functions on the divisor complement. Locally, suitable roots of monomials in these functions are defining coordinates. A convexity argument for tame almost complex structures makes the required functions holomorphic simultaneously.

1. Statement and conventions

A *symplectic simple normal-crossings divisor* will mean a finite union

$$D = \bigcup_{i=1}^{m} D_i$$

of closed embedded symplectic submanifolds of real codimension two, such that every intersection

$$D_I := \bigcap_{i \in I} D_i$$

is transverse and symplectic, and its symplectic orientation agrees with its intersection orientation. Empty intersections are permitted. A labelled hypersurface may be disconnected; this convention does not change the notion, since its connected components are disjoint.

A smooth complex coordinate system means a smooth real coordinate system with its coordinates grouped into complex-valued functions. It is not understood to be holomorphic unless this is explicitly stated.

Theorem 1.1. *Let (M^{2n}, ω) be a closed symplectic manifold, and let $D \subset M$ be a symplectic simple normal-crossings divisor. There exist a smooth path of symplectic forms ω_t, $0 \le t \le 1$, and an ω_1-tame almost complex structure J such that:*

(i) *$\omega_0 = \omega$ and $[\omega_t] = [\omega]$ for every t;*
(ii) *D remains a symplectic simple normal-crossings divisor for every ω_t;*
(iii) *every point of M has smooth complex coordinates $(z_1, \dots, z_n)$ in which*

$$D = \{z_1 \cdots z_r = 0\}$$

for some $0 \le r \le n$, and

$$\mathrm{d}z_j \circ J = i\,\mathrm{d}z_j, \qquad 1 \le j \le r.$$

We use two established results from symplectic geometry: regularization of symplectic normal-crossings divisors, and relative approximately holomorphic hypersurface constructions. We state precisely the consequences needed below, including an explanation of the relative and nonintegral versions of the latter. The remaining argument is elementary.

2. Geometric preliminaries

2.1. Regularization and compatible almost complex structures.

Lemma 2.1. *Let E be a symplectic simple normal-crossings divisor in a closed symplectic manifold (X, Ω). There is a path*

$$\Omega_t = \Omega + \mathrm{d}\mu_t, \qquad \mu_0 = 0,$$

through symplectic forms for which E remains a symplectic simple normal-crossings divisor, such that E admits a symplectic regularization for Ω_1.

Moreover, an Ω_1-compatible almost complex structure K can be chosen to preserve the tangent bundle of every component of E.

Proof. The existence of the exact deformation and the regularization is [3, Theorem 2.13], with the parameter space reduced to a point. Here and below the numbering refers to the version specified in the bibliography.

We explain the last assertion. A regularization supplies, near a point lying on the components indexed by I, a split normal-bundle model

$$N = \bigoplus_{i \in I} N_i \longrightarrow E_I,$$

with the components given by the hyperplane subbundles $v_i = 0$. The symplectic form in this model is

$$\pi^* \Omega_I + \frac{1}{2} \sum_{i \in I} \mathrm{d}(\rho_i \alpha_i),$$

where ρ_i is a squared Hermitian norm and α_i is the angular form of a unitary connection; see [3, Section 2.2]. In particular, along $v_i = 0$, the vertical ith line is the symplectic orthogonal complement of the tangent bundle of that hyperplane.

Using the connection splitting, choose a local Riemannian metric making the horizontal space and the individual vertical lines mutually orthogonal. Along each component this metric makes TE_i orthogonal to $(TE_i)^{\Omega_1}$. Cover X by such neighborhoods and by neighborhoods disjoint from E. A partition of unity gives a global metric h with the same orthogonality property: that property is a linear condition on a metric at a point of E_i.

Define $A \in \mathrm{End}(TX)$ by

$$h(Au, v) = \Omega_1(u, v).$$

Then A is invertible and h-skew-adjoint. Set

$$P = (-A^2)^{1/2}, \qquad K = AP^{-1}.$$

The positive square root is smooth, and A commutes with P. Consequently $K^2 = -\mathrm{id}$, K is h-orthogonal, and

$$\Omega_1(u, Ku) = h(Pu, u) > 0, \qquad \Omega_1(Ku, Kv) = \Omega_1(u, v).$$

Thus K is Ω_1-compatible.

At a point of E_i, the orthogonality property of h implies that A preserves TE_i: if $u \in TE_i$ and $v \in (TE_i)^{\Omega_1}$, then

$$h(Au, v) = \Omega_1(u, v) = 0.$$

Functional calculus now shows that P and K preserve TE_i as well. □

2.2. **Divisor line bundles.** We use the standard classification of smooth complex line bundles by their first Chern class, the equality of the first Chern class with the Euler class of the underlying oriented plane bundle, and the Chern–Weil formula

$$c_1(L)_{\mathbb{R}} = \left[\frac{i}{2\pi} F_\nabla \right].$$

References are [5, Sections 9 and 14] and [4, Chapter XII]. In particular, a complex line bundle with vanishing integral first Chern class is smoothly trivial, not merely trivial after tensoring its cohomology with $\mathbb{R}$.

Lemma 2.2. *Let $Y \subset X$ be a closed embedded codimension-two submanifold with a complex structure on its normal bundle N_Y. There are a smooth complex line bundle $L(Y) \to X$ and a transverse section σ_Y such that*

$$\sigma_Y^{-1}(0) = Y, \qquad c_1(L(Y)) = \mathrm{PD}[Y].$$

The intrinsic derivative along its zero set identifies

$$\mathrm{d}\sigma_Y : TX|_Y/TY \longrightarrow L(Y)|_Y$$

with the identity of N_Y.

In particular, if an almost complex structure K preserves TY and the normal complex structure is induced by K, then

$$\mathrm{d}\sigma_Y \circ K = i\,\mathrm{d}\sigma_Y \quad \text{along } Y.$$

Proof. Identify a tubular neighborhood T of Y with a disk neighborhood of the zero section of N_Y. On T take the bundle π^*N_Y and its tautological section $v \mapsto v$. On $X \setminus Y$ take the trivial line bundle with section 1. Over $T \setminus Y$, glue these bundles by identifying 1 with the nonzero vector v. The sections glue to σ_Y. Its derivative on the normal bundle is the identity.

The zero locus of a transverse section represents the Euler class of the underlying oriented plane bundle, hence the first Chern class of this complex line bundle. This gives the asserted Poincaré duality formula, with the orientation determined by the specified normal complex structure. □

2.3. **The relative hypersurface input.** For a sequence of tensor fields, the notation $O_{C^\infty}(1)$ means boundedness in every fixed derivative norm, with respect to fixed geometric data on the base.

Lemma 2.3 (Relative approximately holomorphic hypersurfaces)**.** *Let (X, Ω, K) be a closed symplectic manifold with K Ω-compatible. Let $Y_1, \dots, Y_q$ be a finite collection of closed K-invariant submanifolds, including X itself if desired. Let L_k be Hermitian line bundles with unitary connections satisfying*

$$F_{L_k} = -ik\Omega + O_{C^\infty}(1).$$

For every sufficiently large k, there is a smooth section s_k of L_k which is transverse to zero on X and whose restriction to every Y_a is transverse to zero. All the resulting zero sets in X and in the Y_a are symplectic, with their complex-normal zero-set orientations.

Proof. We recall the precise approximately holomorphic input and its relative application. Put $g = \Omega(\,\cdot\,, K\,\cdot\,)$ and $g_k = kg$. The curvature hypothesis makes L_k an asymptotically very ample sequence in the sense of [2, Definition 2.1]: its curvature is positive of order k, its $(0,2)$-part is bounded, and its curvature derivatives are $O(k)$.

Estimated transversality gives approximately holomorphic sections, with arbitrarily small approximately holomorphic perturbations available, for which the restrictions in question are uniformly transverse. The relevant general construction is [2, Theorem 1.1 and Section 4]; the restriction-to-a-prescribed-submanifold version is the main relative transversality theorem of [6].

For clarity, the passage to the particular formulation above requires no extension theorem for holomorphic sections. Use the ambient Gaussian peak sections centered at points of Y_a, supplied by [2, Lemma 2.3]. Their restrictions to the K-invariant submanifold Y_a have the same local approximately holomorphic estimates and nonvanishing lower bounds. Thus the local perturbations for transversality on Y_a are restrictions of ambient perturbations. The Gaussian-decay globalization argument applies with centers on Y_a. Apply this successively to the finite list of submanifolds, taking each perturbation sufficiently small to preserve the previous uniform transversality bounds. The openness and successive-perturbation argument is the one used in [1, proof of Theorem 2].

More explicitly, the resulting sections satisfy, for constants independent of k,

$$\|s_k\|_{C^2(g_k)} \leq C, \qquad \|\bar{\partial}_K s_k\|_{C^1(g_k)} \leq Ck^{-1/2},$$

and their derivatives, restricted to each TY_a, have right inverses of uniformly bounded norm wherever the restricted section is sufficiently small. This also explains why the construction applies to the above general sequence L_k, rather than only to tensor powers of one line bundle.

At a zero in Y_a, the derivative of $s_k|_{Y_a}$ is therefore a uniformly surjective real-linear map whose antilinear part tends to zero. Its kernel is close to the kernel of a surjective complex-linear map. For large k this kernel is symplectic. The homotopy which removes the small antilinear part stays surjective, so its zero-set orientation is the symplectic orientation. The same argument applies on X. For a zero-dimensional Y_a, transversality means that the section has no zero there. □

Remark 2.4. Suppose E is a symplectic simple normal-crossings divisor and K is compatible with Ω and preserves all its components. Apply Lemma 2.3 to all the nonempty intersections E_I, including $E_\varnothing = X$. The new zero set H then enlarges E to a symplectic simple normal-crossings divisor.

Indeed, transversality on E_I is exactly the additional transversality required for $E_I \cap H$. The orientation assertion in the lemma gives the required agreement of intersection and symplectic orientations.

3. A cohomologically redundant enlargement

The purpose of the auxiliary hypersurfaces is purely cohomological. Importantly, their construction does not require the original symplectic class to be rational.

Lemma 3.1. *Suppose E is a symplectic simple normal-crossings divisor in (X, Ω) and there is an Ω-compatible almost complex structure preserving its components. Let*

$$W \subsetneq H^2(X; \mathbb{Q})$$

be a proper rational vector subspace. There is a closed embedded symplectic hypersurface H such that $E \cup H$ is a symplectic simple normal-crossings divisor and

$$\mathrm{PD}[H] \notin W.$$

Proof. Choose integral classes $\xi_1, \ldots, \xi_b$ whose images form a basis of $H^2(X; \mathbb{Z})$ modulo torsion. Choose closed real forms β_j representing ξ_j, and write

$$[\Omega] = 2\pi \sum_{j=1}^{b} \lambda_j \xi_j \quad \text{in } H^2(X; \mathbb{R}).$$

There is a real one-form η such that

$$\Omega = 2\pi \sum_{j=1}^{b} \lambda_j \beta_j + \mathrm{d}\eta.$$

Choose Hermitian line bundles B_j with $c_1(B_j) = \xi_j$ and unitary connections of curvature $-2\pi i \beta_j$. Such connections exist: start with any unitary connection and correct it by an imaginary-valued one-form, using the Chern–Weil formula.

Let $m_j(k)$ be an integer nearest to $k\lambda_j$, and set

$$L_k = \bigotimes_{j=1}^{b} B_j^{\otimes m_j(k)}.$$

Equip L_k with the tensor-product connection, modified by the imaginary-valued one-form $-ik\eta$. Its curvature is

$$\begin{aligned} F_{L_k} &= -2\pi i \sum_j m_j(k)\beta_j - ik\,\mathrm{d}\eta \\ &= -ik\Omega - 2\pi i \sum_j \big(m_j(k) - k\lambda_j\big)\beta_j \\ &= -ik\Omega + O_{C^\infty}(1). \end{aligned}$$

Write $h_k = c_1(L_k)$.

Choose an integral class a whose rational image is not in W, and a Hermitian line bundle A with $c_1(A) = a$, equipped with any unitary connection. The sequence $L_k \otimes A$ satisfies the same curvature estimate.

By Lemma 2.3 and Remark 2.4, for all sufficiently large k both classes

$$h_k, \qquad h_k + a$$

are represented by hypersurfaces which can separately be added to E as symplectic simple normal-crossings components. They cannot both belong to W, since their difference is a. Choose the hypersurface corresponding to a class outside W. □

Lemma 3.2. *Starting with (M^{2n}, ω, D), one can make a finite sequence of the exact deformations allowed in Theorem 1.1 and add finitely many auxiliary hypersurfaces to obtain a symplectic simple normal-crossings divisor*

$$E = \bigcup_{i=1}^{N} E_i, \qquad D \subset E,$$

in (M, Ω) with the following properties:

(a) *there is an Ω-compatible almost complex structure preserving every E_i;*
(b) *for every $I \subset \{1, \dots, N\}$ with $|I| \le n$, the classes*

$$\{\mathrm{PD}[E_j] : j \notin I\}$$

span $H^2(M; \mathbb{Q})$.

Throughout the deformations, the original divisor D is fixed as a subset and remains symplectic simple normal crossings.

Proof. Set $b = \dim H^2(M; \mathbb{Q})$. First apply Lemma 2.1 to the current divisor. We construct $n + 1$ disjoint groups of auxiliary labels, each group containing b hypersurfaces whose classes form a basis of $H^2(M; \mathbb{Q})$.

Within one group, suppose $\ell < b$ linearly independent classes have already been obtained. Apply Lemma 3.1 with W their span. Add the resulting hypersurface to the current divisor. Apply Lemma 2.1 to this enlarged divisor before proceeding to the next step. Its deformation is exact and preserves all the components already constructed. This procedure constructs the required basis group, and repeating it constructs all $n + 1$ groups.

At the end, the compatible almost complex structure from the last regularization gives (a). Deleting at most n labels can meet at most n of the $n + 1$ groups, so at least one complete basis group remains. This proves (b). If $b = 0$, the spanning condition is vacuous and no auxiliary components are necessary.

All forms encountered have cohomology class $[\omega]$. The finitely many deformation paths can be concatenated smoothly by making their parametrizations constant near their endpoints. □

3.1. **An integral relation matrix.**

Lemma 3.3. *Let $c_1, \dots, c_N \in H^2(M;\mathbb{Z})$. Suppose that, after deleting any set of at most n indices, the remaining rational classes still span the rational span of all the c_i. There is a symmetric integer matrix*

$$B = (b_{ij})_{1\le i,j\le N}$$

such that

$$\sum_{j=1}^{N} b_{ij}c_j = 0 \quad \text{in } H^2(M;\mathbb{Z}), \qquad 1 \le i \le N,$$

and every nonempty principal submatrix

$$B_I = (b_{ij})_{i,j\in I}, \qquad |I| \le n,$$

is positive definite.

Proof. Choose a rational basis of the span of the classes and let C be the matrix whose jth column is the coordinate vector of c_j. Thus C has full row rank. The orthogonal projection onto its kernel is the rational symmetric matrix

$$P = I_N - C^{\mathsf{T}}(CC^{\mathsf{T}})^{-1}C.$$

If the rational span is zero, use $P = I_N$.

We claim that P_I is positive definite when $|I| \le n$. If $v \in \mathbb{R}^N$ is supported on I and $v^{\mathsf{T}}Pv = 0$, then $Pv = 0$, since P is an orthogonal projection. Hence $v = C^{\mathsf{T}}u$ for some u. For $j \notin I$ the jth entry of v is zero, so u is orthogonal to every column of C outside I. Those columns span the whole column space, by hypothesis. Thus $u = 0$ and $v = 0$, proving the claim.

Multiply P by a positive integer to clear its denominators. Every resulting row gives an integral combination of the c_j which vanishes rationally, hence is torsion. The torsion subgroup of $H^2(M;\mathbb{Z})$ is finite. Multiplying once more by its exponent kills all these torsion classes. The resulting integer matrix B has the asserted properties. □

4. Gluing tame structures while preserving holomorphic functions

The following observation is the gluing mechanism in the proof. It is useful that holomorphicity of a fixed function becomes an affine-linear condition in suitable coordinates on the space of tame almost complex structures.

Lemma 4.1. *Let (X,Ω) be a symplectic manifold. Suppose $\{U_\alpha\}$ is a locally finite open cover, J_α is an Ω-tame almost complex structure on U_α, and $\{\rho_\alpha\}$ is a subordinate smooth partition of unity.*

There is a smooth Ω-tame almost complex structure J with the following pointwise property. If $\lambda : T_xX \to \mathbb{C}$ is a complex-valued real-linear covector and

$$\lambda \circ J_\alpha(x) = i\lambda$$

for every α with $\rho_\alpha(x) > 0$, then

$$\lambda \circ J(x) = i\lambda.$$

Proof. Choose an Ω-compatible reference structure J_*. Write

$$TX \otimes_{\mathbb{R}} \mathbb{C} = V_- \oplus V_+,$$

where V_- and V_+ are the $-i$ and $+i$ eigenbundles of J_*. On the complexified tangent bundle consider the Hermitian form

$$H(v,w) = i\Omega_{\mathbb{C}}(v,\overline{w}).$$

It is positive definite on V_-, negative definite on V_+, and these two subbundles are H-orthogonal.

For any Ω-tame almost complex structure J', its $-i$ eigenspace is H-positive. Indeed, its vectors have the form $u + iJ'u$, and

$$H(u + iJ'u, u + iJ'u) = 2\Omega(u, J'u) > 0$$

when $u \neq 0$. Projection to V_- therefore identifies this eigenspace with the graph of a complex-linear map

$$A' : V_- \longrightarrow V_+, \qquad \|A'\| < 1,$$

where the norms come from H on V_- and $-H$ on V_+.

Conversely, the graph of any such contraction is H-positive. Its conjugate is H-negative, so the two graphs are complementary. They define a real almost complex structure, which is tame by the same calculation.

Let A_α represent J_α in these graph coordinates, and set

$$A = \sum_\alpha \rho_\alpha A_\alpha.$$

The products are extended by zero outside their respective domains. This is smooth and satisfies

$$\|A\| \leq \sum_\alpha \rho_\alpha \|A_\alpha\| < 1.$$

Let J be the tame structure defined by its graph.

The condition $\lambda \circ J_\alpha = i\lambda$ is equivalent to the complex-linear extension of λ vanishing on the $-i$ eigenspace, or

$$\lambda(v) + \lambda(A_\alpha v) = 0 \qquad (v \in V_-).$$

This is affine-linear in A_α. Taking the weighted sum proves the corresponding identity for A, and hence the claimed property of J. □

5. Monomial functions and holomorphic normal coordinates

Proposition 5.1. *Let $E = \bigcup_{i=1}^N E_i$ be a symplectic simple normal-crossings divisor in a closed symplectic manifold (X^{2n}, Ω). Assume:*

(a) *there is an Ω-compatible almost complex structure K preserving every E_i;*
(b) *the classes $c_i = \mathrm{PD}[E_i]$ satisfy the hypothesis of Lemma 3.3.*

Then there is an Ω-tame almost complex structure J for which, near every point, all the local branches of E have simultaneous J-holomorphic defining coordinates.

Proof. We divide the argument into four steps.

Step 1: global functions on the complement. For each E_i, use the normal complex structure induced by K and Lemma 2.2 to obtain a complex line bundle L_i and a section σ_i satisfying

$$\sigma_i^{-1}(0) = E_i, \qquad c_1(L_i) = c_i, \qquad \mathrm{d}\sigma_i \circ K = i\,\mathrm{d}\sigma_i \quad \text{along } E_i.$$

Choose the matrix B from Lemma 3.3. Each line bundle

$$P_i := \bigotimes_{j=1}^N L_j^{\otimes b_{ij}}$$

has zero integral first Chern class, and is therefore smoothly trivial. Choose a complex-linear trivialization

$$\tau_i : P_i \longrightarrow X \times \mathbb{C}.$$

A negative tensor power denotes a power of the dual line.

On $X \setminus E$, all the sections σ_j are nonzero. They therefore define smooth functions

$$f_i = \tau_i \left(\bigotimes_{j=1}^{N} \sigma_j^{\otimes b_{ij}} \right) : X \setminus E \longrightarrow \mathbb{C}^*.$$

For negative exponents, the inverse section of the dual line is used. No holomorphicity is asserted yet.

Step 2: local simultaneous monomialization. Fix $p \in E$, and put

$$I = I(p) := \{i : p \in E_i\}.$$

Then $1 \leq |I| \leq n$. Take a sufficiently small neighborhood U of p which meets no component E_j with $j \notin I$, and trivialize all the L_j there. Write

$$\sigma_j = g_j e_j$$

in these frames. For $j \in I$, put $z_j = g_j$. Transversality of the divisor means that the map $(z_j)_{j \in I}$ is a submersion near p. Moreover,

$$\mathrm{d}z_j(p) \circ K(p) = i\, \mathrm{d}z_j(p), \qquad j \in I.$$

Complete these functions to smooth complex coordinates

$$\big((z_j)_{j \in I}, y_1, \ldots, y_{n-|I|}\big)$$

whose differential at p is complex-linear for $K(p)$. To do this, extend the independent complex covectors $\mathrm{d}z_j(p)$ to a complex basis of the cotangent space, choose smooth functions with these additional differentials, and apply the inverse function theorem.

For $i \in I$, the definition of f_i gives

$$f_i = a_i \prod_{j \in I} z_j^{b_{ij}} \quad \text{on } U \setminus E,$$

where each $a_i : U \to \mathbb{C}^*$ is smooth: all factors belonging to indices outside I are nonvanishing and are absorbed into a_i.

Shrink U so that a_i admits a smooth logarithm h_i for each $i \in I$. Since B_I is invertible, define

$$u_j = \sum_{i \in I} (B_I^{-1})_{ji} h_i, \qquad w_j = z_j e^{u_j}, \qquad j \in I.$$

Then

$$f_i = \prod_{j \in I} w_j^{b_{ij}}, \qquad i \in I, \quad \text{on } U \setminus E. \tag{1}$$

Also,

$$\{w_j = 0\} = E_j \cap U, \qquad \mathrm{d}w_j(p) = e^{u_j(p)} \mathrm{d}z_j(p).$$

Thus, after shrinking once more,

$$\big((w_j)_{j \in I}, y_1, \ldots, y_{n-|I|}\big)$$

is a smooth complex coordinate system whose differential at p is complex-linear for $K(p)$.

Let J_p be the pullback of the standard complex structure in these coordinates. It agrees with K at p, so it is Ω-tame on a sufficiently small neighborhood U_p. Every f_i with $i \in I(p)$ is J_p-holomorphic on $U_p \setminus E$, by (1).

At a point outside E, choose any local Ω-tame almost complex structure and give the corresponding neighborhood the empty label set. We have therefore constructed an open cover by neighborhoods U_α, with label sets I_α, such that:

(1) $U_\alpha \cap E_i = \varnothing$ if $i \notin I_\alpha$;
(2) J_α is Ω-tame on U_α;

(3) f_i is J_α-holomorphic on $U_\alpha \setminus E$ whenever $i \in I_\alpha$.

Take a finite such cover, using compactness of X.

Step 3: gluing. Choose a smooth partition of unity $\{\rho_\alpha\}$ subordinate to this finite cover, with $\operatorname{supp}\rho_\alpha \subset U_\alpha$. Apply Lemma 4.1 to obtain an Ω-tame almost complex structure J.

For each i, define the open neighborhood

$$V_i = X \setminus \bigcup_{\alpha:\, i\notin I_\alpha} \operatorname{supp}\rho_\alpha.$$

Property (1) above shows that $E_i \subset V_i$. If $x \in V_i \setminus E$, every index with $\rho_\alpha(x) > 0$ has $i \in I_\alpha$. Thus every contributing J_α satisfies

$$\mathrm{d}f_i(x) \circ J_\alpha(x) = i\,\mathrm{d}f_i(x).$$

The gluing lemma gives

$$\mathrm{d}f_i \circ J = i\,\mathrm{d}f_i \quad \text{on } V_i \setminus E. \tag{2}$$

Step 4: recovery of the defining coordinates. Return to $p \in E$, set $I = I(p)$, and use the local coordinates constructed in Step 2. Restrict their neighborhood to

$$U \cap \bigcap_{i\in I} V_i.$$

On its complement of E, all the functions f_i and w_j appearing in (1) are nonzero. Logarithmic differentiation yields

$$\frac{\mathrm{d}f_i}{f_i} = \sum_{j\in I} b_{ij}\frac{\mathrm{d}w_j}{w_j}, \qquad i \in I.$$

By (2), the covectors on the left are complex-linear for J. Since B_I is invertible, the same is true of every $\mathrm{d}w_j/w_j$, and hence of every $\mathrm{d}w_j$:

$$\mathrm{d}w_j \circ J = i\,\mathrm{d}w_j \quad \text{on } U \setminus E.$$

Both sides are smooth on all of U. The complement of the finite union of codimension-two submanifolds E is dense. Consequently this identity holds throughout U.

The functions w_j, $j \in I$, are therefore simultaneous J-holomorphic defining coordinates, and

$$E \cap U = \left\{ \prod_{j\in I} w_j = 0 \right\}.$$

This proves the proposition. □

6. Proof of the main theorem

Proof of Theorem 1.1. If D is empty, choose any ω-compatible almost complex structure and take $r = 0$ everywhere. The zero-dimensional case is likewise immediate. Assume henceforth that $n \geq 1$ and $D \neq \varnothing$.

Apply Lemma 3.2. It produces an endpoint symplectic form Ω, connected to ω by a smooth path of cohomologous symplectic forms preserving the original divisor, and an enlarged divisor

$$E = D \cup E_{\mathrm{aux}}$$

satisfying the hypotheses of Proposition 5.1. Let J be the Ω-tame almost complex structure provided by that proposition.

At a point $p \in D$, the proposition gives smooth complex coordinates in which every branch of the enlarged divisor through p is a coordinate hyperplane, with its defining coordinate J-holomorphic. Retain as the first $z_1, \ldots, z_r$ precisely the coordinates corresponding to the original components of D through p. Use the remaining coordinates, including any auxiliary defining coordinates, to

complete the smooth complex coordinate system. Shrinking the neighborhood to avoid all other original components gives

$$D = \{z_1 \cdots z_r = 0\}, \qquad \mathrm{d}z_j \circ J = i\,\mathrm{d}z_j \quad (1 \le j \le r).$$

At a point outside D, choose a neighborhood disjoint from D and take $r = 0$.

Finally, set $\omega_1 = \Omega$ and use the concatenated path from Lemma 3.2. It has all the properties asserted in the theorem. □

Remark 6.1. The argument does not confuse holomorphic defining functions with the weaker condition that the Nijenhuis tensor have tangential image along the divisor. Holomorphicity here holds on actual open neighborhoods. It follows from the global functions f_i, the gluing identity (2), and the invertibility of the principal matrices B_I.

Remark 6.2. Neither the deformation nor the auxiliary construction changes the cohomology class of the symplectic form. The rounding construction in Lemma 3.1 is what permits approximately holomorphic hypersurfaces to be used without assuming that $[\omega]$ is rational. Separately, the torsion-killing step in Lemma 3.3 ensures that the tensor-product line bundles used to define the functions f_i are genuinely trivial.

References

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